

CHANGES IN PRODUCTS AND SERVICES QUALITY
IN SPATIAL MARKETS; INFORMATION
TRANSMISSION AND LOCATION CHOICES

ANAÏS GALDIN

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ADVISERS: EDUARDO MORALES AND STEPHEN J. REDDING

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Abstract

To be edited...

The first two chapters study changes in product and service characteristics in globalized markets, in which capturing information about quality among a thick range of suppliers and workers is complicated. The third chapter studies transmission of information, but at the scale of the U.S. for years 1870-1940. This historical context offers a set up where the transmission of information in space was more limited.

Chapter 1 studies the lack of resilience of global pharmaceutical supply chains since 2009 and the extent to which the offshoring of manufacturing plants to Asia is driving persistent drug shortages in the United States. I build a novel dataset that, for the first time, allows to determine the exact manufacturing location of drug products sold in the U.S. The data solve the traceability issue in the market and enable me to document new empirical evidence on the causal impact of offshoring on decreased pharmaceutical supply chain resilience. Compared to their U.S.-made counterparts, drugs offshored to Asia are 56% more likely to face shortages, which last 130 days longer on average. To unveil the incentive mechanisms behind supply disruption, I develop and estimate a structural model of spatial location choices and global drug procurement. Suppliers trade off supply reliability and cost-cutting, which endogenously generates shortages. The fundamental problem with the market lies in its inability to capture and reward supply reliability under the current procurement system. Offshoring serves as a lever for drug manufacturers to exploit laxer regulatory oversight and target higher disruption risks than permissible in the U.S., in exchange for lower production costs. The resulting increase in average supply disruption risk, coupled with short-term pricing and capacity rigidities in generic injectable drug markets, translates into shortages. I use the estimated model to analyze two types of counterfactuals: reshoring production and modifying procurement contracts. While reshoring does not achieve the first best, enforcing failure-to-supply clauses in contracts effectively rewards supply reliability through higher

producer margins. The net consumer welfare gains of reducing shortages may be large and are not offset by the associated increase in drug prices.

Chapter 2, joint with Jesse Silbert, investigates the trade-off agents face when providing information to potential matches in two-sided markets. Specifically, we explore how, in online labor markets, market congestion can lead to under-provision of information from global workers to employers and ultimately negatively impact match efficiency. We develop a model of equilibrium information provision characterized by attention-constrained employers and workers who provide match-specific costly information to their potential employers. The model shows that congestion forms a binding attention constraint, leading to under-provision of signaling effort by workers in equilibrium. We then provide empirical evidence from a major international task-based online labor market that supports our theoretical model, showing that congestion leads to hasty and generic job applications and that the market is characterized by low job-filling rates. Our model and setting lay the groundwork needed to quantify the loss of efficiency that arises due to information under-provision in congested markets.

Chapter 3, joint with Quan Le, investigates the role of newspapers in the migration decisions of rural individuals in the 1870-1940 United States. Contributing to the urban and health economics literature, we offer new insights about how information, specifically newspaper portrayals of public health advancements like water filtration and sewage systems, shapes rural-urban migration patterns. We show that access to information about urban health conditions through newspapers played a crucial role in encouraging rural individuals to move to cities over the turn of the 19th century. By exploiting a novel linkage between full-count U.S. census data and a unique historical newspaper dataset at the county level, which leverages text-mining and Natural Language Processing (NLP) algorithms, we provide novel evidence that rural individuals responded to newspaper narratives on public health investments and typhoid occurrences. Rural migrants with access to information about public

health investments migrated in higher proportion to sanitation-adopting cities compared to rural migrants with no access to such information, and avoided cities affected by pandemics more than their non-informed counterparts. This finding is consistent with the literature that suggests that such investments significantly reduced mortality from waterborne diseases (typhoid and diphtheria) and improved quality of life, making cities more attractive places to live. As policymakers consider strategies to revitalize urban areas in the post-pandemic era, our study highlights the potential importance of information dissemination in promoting urban growth and development.

Acknowledgements

Coming Soon....

Dedication. Coming Soon....

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Chapter 1

Resilience of Global Supply Chains and Generic Drug Shortages

Chapter 2

Equilibrium Provision of Information in Two-Sided Markets: Evidence from an Online Job Platform

Chapter 3

Urban Migration, Public Health Amenities, and Local Newspapers in 1870-1940 U.S.